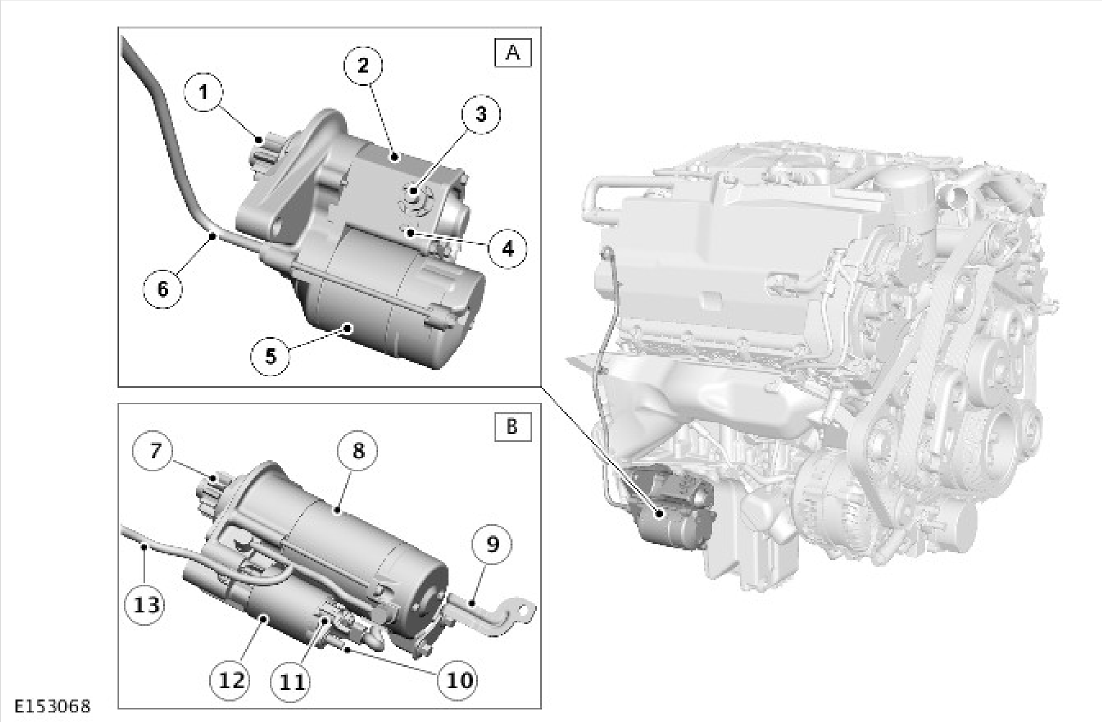
PUBLISHED: 28-MAY-2014
2014.0 RANGE ROVER (LG), 303-06

STARTING SYSTEM - V8 N/A 5.0L PETROL/V8 S/C 5.0L PETROL

COMPONENT LOCATION

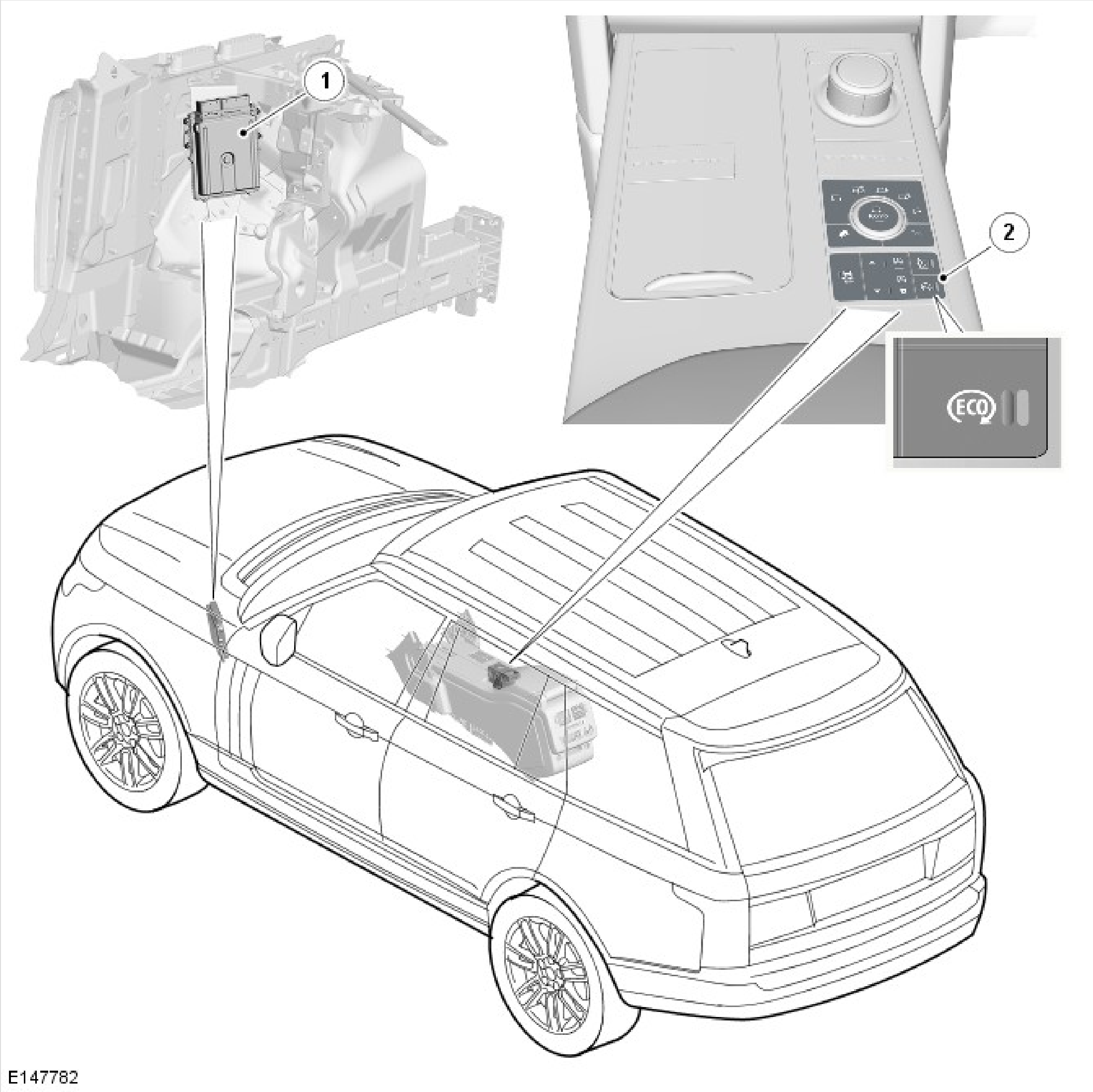
Starter Motor



ITEM	DESCRIPTION
A	Non TSS (tandem solenoid starter) Motor
B	TSS (tandem solenoid starter) Motor
1	Pinion Gear
2	Solenoid
3	Motor Electrical Connection
4	Solenoid Electrical Connection
5	Motor
6	Breather Tube
7	Pinion Gear
8	Motor
9	Support Bracket

10	Motor Electrical Connection
11	Solenoid Electrical Connection
12	Solenoid
13	Heatshield
14	Breather Tube

Stop/Start System



ITEM	DESCRIPTION
1	ECM (engine control module)
2	ECO Switch

OVERVIEW

The starting system consists of an electric starter motor that engages with the ring gear on the crankshaft drive plate. Operation of the starter motor is controlled by the ECM (engine control module) .

In some markets the starting system incorporates a stop/start system. The system is controlled by the ECM using CAN (controller area network) messages and signals from other system components and modules to determine the correct conditions for system operation. The stop/start system automatically stops and re-starts the engine when appropriate conditions are met. This reduces the amount of time the engine is running at idle speed which improves economy and reduces engine emissions.

The stop/start system is automatically enabled each time the ignition is switched on. The driver can disable the system using an ECO switch.

The stop/start system stops the engine when it is not needed, this is known as ECO stop. The engine will restart automatically when vehicle parameters are met and the driver releases the brake pedal, this is known as ECO start.

To accommodate the increased electrical loads, vehicles fitted with the stop/start system are equipped with two batteries.

DESCRIPTION

Vehicles without the stop/start system use a conventional starter motor. Vehicles with the stop/start system use a TSS (tandem solenoid starter) motor. The TSS motor can restart the engine if it is still rotating, allowing the engine to be started quickly in any situation.

The starter motor is located on the rear right side of the oil pan. The rear of the starter motor is attached to a support bracket on the oil pan. The starter motor incorporates a snorkel breather pipe to assist with ventilation and sealing, during water submergence.

Vehicles without Stop/Start

The starter motor is rated at 1.4 kW and comprises a series wound motor, an overrunning clutch and pinion, and a solenoid.

Power to operate the motor is supplied from the 450 A megafuse in BJB2 (battery junction box 2). Power to operate the solenoid is supplied from the starter relay in the EJB (engine junction box) .

Vehicles with Stop/Start

The TSS motor is rated at 2.0 kW and comprises a series wound motor, an overrunning clutch and pinion, a pinion gear shift solenoid and a motor rotation solenoid.

Power to operate the motor is supplied from the 450A megafuse in BJB2. Power to operate the solenoids is supplied from the starter motor relay and the starter pinion relay in the EJB .

ECO SWITCH

On vehicles with stop/start, the ECO switch is integrated into the Terrain Response® switchpack in the floor console. An amber LED (light emitting diode) in the ECO switch remains illuminated while the stop/start system is active. Selection of the ECO switch is transmitted from the Terrain Response® switchpack to the ECM on the high speed CAN chassis, and high speed CAN powertrain bus via the GWM (gateway module).

WARNING INDICATOR

On vehicles with stop/start, an ECO warning indicator is located in the IC (instrument cluster). The warning indicator comes on when the engine stops during a stop/start cycle, then goes off when the engine restarts. The warning indicator is controlled by a high speed CAN powertrain bus message from the ECM .

OPERATION

VEHICLES WITHOUT STOP/START

When the engine stop/start switch is pressed, if the keyless vehicle module detects a valid smart key in the vehicle the CJB (central junction box) transmits a hardwired crank request to the ECM .
For additional information, refer to: Anti-Theft - Passive (419-01 Anti-Theft - Passive, Description and Operation).

When the ECM receives a crank request from the CJB it energizes the starter relay, provided the TCS (transmission control switch) is in P (park) or N (neutral) and the brake pedal is pressed. The energized starter relay supplies battery power to the starter solenoid, which energizes and causes the pinion gear to engage with the ring gear. When the starter solenoid is energized it also closes high-current contacts, which connects battery power from BJB2 (battery junction box 2) to the motor to turn the engine.

VEHICLES WITH STOP/START

At the beginning of a drive cycle, when the engine stop/start switch is pressed the starting system operates in the same way as on vehicles without stop/start, up to the point that the ECM energizes the starter relay. At that point, on the stop/start system, the ECM energizes both the starter pinion relay and the starter motor relay. The energized starter motor relay supplies battery power to the pinion gear shift solenoid in the starter, which energizes and causes the pinion gear to engage with the ring gear. The energized starter pinion relay supplies battery power to the motor rotation solenoid in the starter, which energizes and closes high-current contacts, which connects battery power from BJB2 to the motor to turn the engine.

During a drive cycle, with the stop/start system enabled and the TCS (transmission control switch) in D (drive) or S (sport), the ECM performs an ECO stop when one of the following occurs:

- The vehicle stops from a speed greater than 4 km/h (2.5 mph) and sufficient brake pressure is applied to the brake pedal to ensure the vehicle is stationary
- The vehicle is stationary and N is selected on the TCS.

NOTE:

On a gradient an ECO stop will only occur if the driver applies a minimum of 7 bar (101.5 lbf/in²)+ / - 3 bar (43.5 lbf/in²) brake pressure and will not occur at all on gradients about in excess of 1 in 4.34 (23%).

The green ECO warning indicator in the IC (instrument cluster) illuminates to advise that the engine has shut down. Other warning indicators normally associated with an engine shut down, for example the ignition and low oil pressure warning indicators, are suppressed so will not illuminate during an ECO stop.

The following conditions will prevent an ECO stop:

- A steering wheel paddle switch has been used to select a gear (transmission Commandshift mode).
- The ambient air temperature is outside the range of 0 to 40 °C (32 to 104 °F).
- The engine coolant temperature is outside the range of 20 to 110 °C (68 to 230 °F).

- Brake vacuum is less than the threshold value.
- The pressure altitude derived from the barometric pressure sensor in the ECM is more than approximately 2800 m (9150 feet).
- The hood is open.
- The driver door is open.
- The driver seatbelt is not fastened.
- High demand from the climate control system requires the engine to be running, for example windshield demist selected.
- The battery charge is low.
- A Terrain Response® special program is active.
- Hill descent control is active.
- ECO stop/start is disabled using the ECO switch.

While the vehicle is stationary, if a condition that prevented an ECO stop is removed the ECM automatically performs an ECO stop. The ECM automatically restarts the engine when one of the following occurs:

NOTE:

When the brake pedal is released during an ECO restart the ABS (anti-lock brake system) control module briefly applies the brakes to hold the vehicle stationary until the engine has restarted.

- The brake pedal is released with D or S selected on the TCS.
- The accelerator pedal is pressed.
- A paddle switch is used to select a gear.
- R (reverse) is selected on the TCS.
- Climate control system demand increases.
- The vehicle speed exceeds approximately 1 km/h (0.5 mph).
- The battery charge becomes low.
- Brake vacuum reduces below the threshold value.
- ECO stop/start is disabled using the ECO switch.

The following will prevent an ECO restart, and require use of the engine stop/start switch to restart the engine:

- The hood is open.
- The driver door is open.
- The driver seatbelt is not fastened.

STOP/START STRATEGY

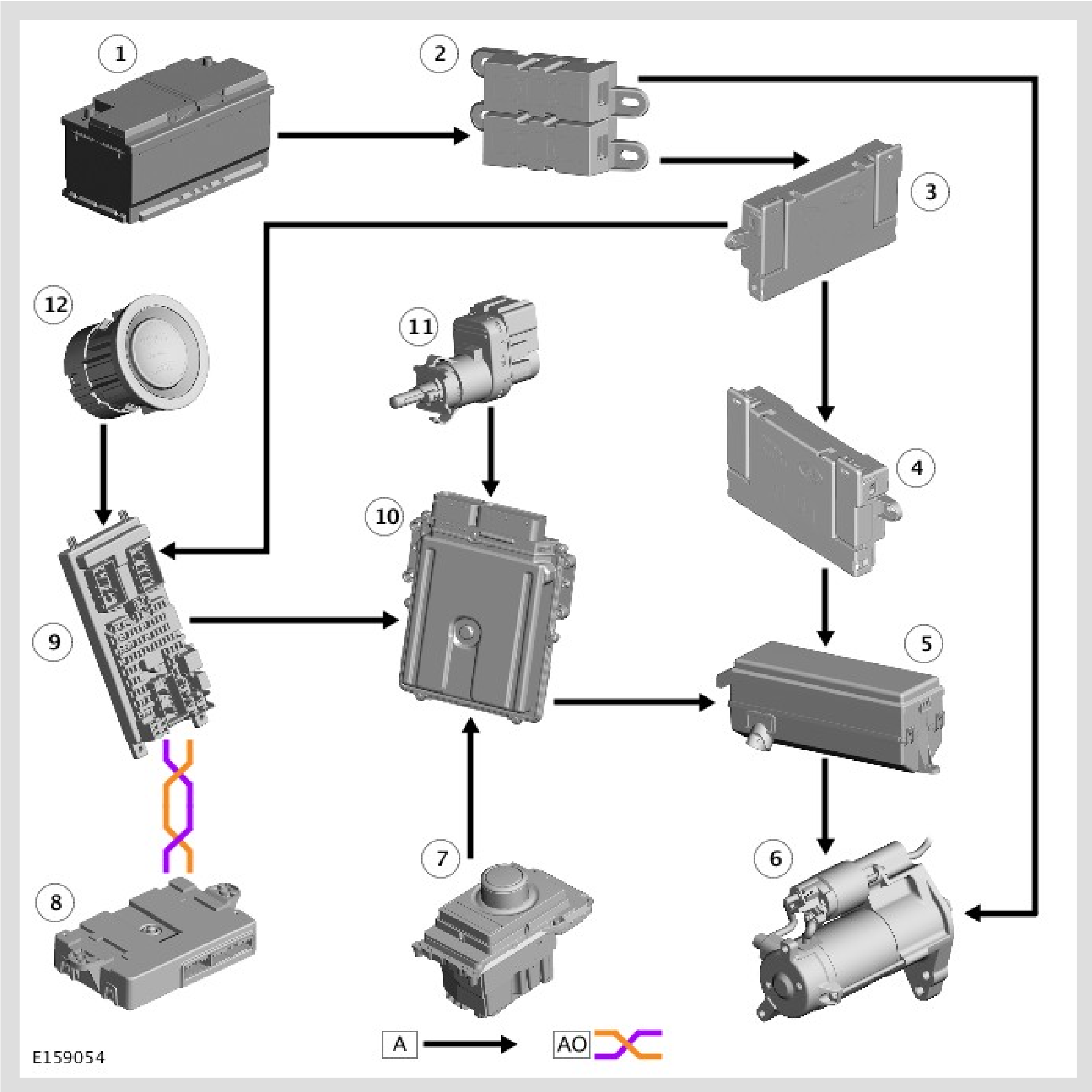
Operating the motor and the pinion gear individually allows meshing of the pinion gear with the drive plate ring gear, even if the engine is still turning. This allows instant restarts under all conditions.

Engine speed 250 rev/min or more: When an ECO stop is initiated and the ECM then detects an engine restart parameter, it operates the starter motor to first accelerate the pinion gear up to the ring gear speed, then engages the pinion gear with the ring gear and energizes the motor to rotate the engine up to starting speed. The ECM also activates the fueling and engine systems to restart the engine.

Engine speed less than 250 rev/min: When an ECO stop is initiated and the ECM then detects an engine restart parameter, it operates the starter motor to first engage the pinion gear with the ring gear, then energizes the motor to rotate the engine up to starting speed. The ECM also activates the fueling and engine systems to restart the engine.

CONTROL DIAGRAM

Vehicles without Stop/Start

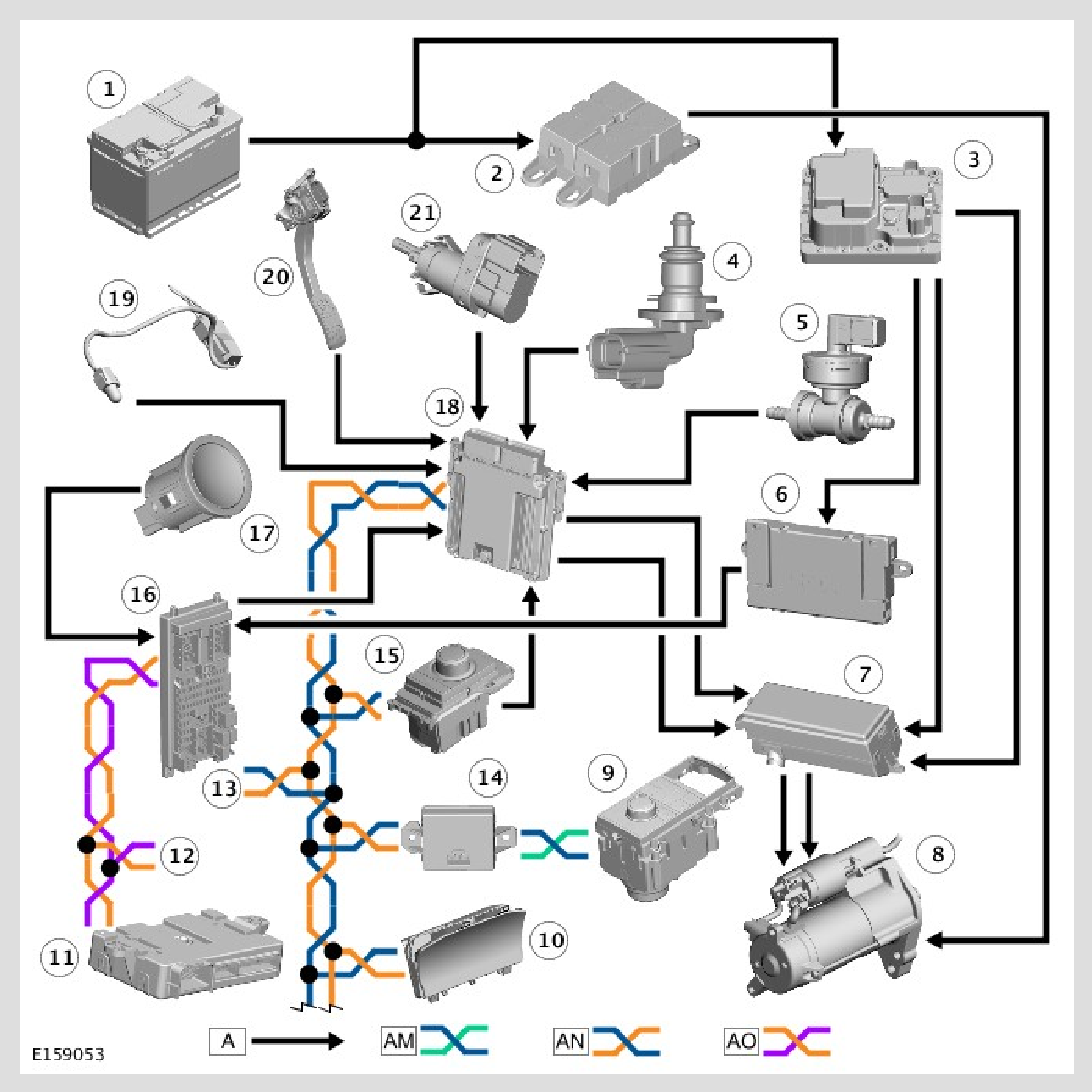


A = HARDWIRED, AO = MEDIUM SPEED CAN BODY BUS.

ITEM	DESCRIPTION
1	Battery
2	BJB2 (battery junction box 2)
3	BJB (battery junction box)
4	AJB (auxiliary junction box)

5	EJB (engine junction box)
6	Starter Motor
7	TCS (transmission control switch)
8	KVM (keyless vehicle module)
9	CJB (central junction box)
10	ECM (engine control module)
11	Brake Pedal Switch
12	Stop/Start Switch

Vehicles with Stop/Start



A = HARDWIRED; AM = HIGH SPEED CAN CHASSIS BUS; AN = HIGH SPEED CAN POWERTRAIN BUS; AO = MEDIUM SPEED CAN BODY BUS

ITEM	DESCRIPTION

1	Battery
2	BJB2 (battery junction box 2)
3	Power Supply Distribution Box
4	ECT (engine coolant temperature) Sensor
5	Brake Vacuum Sensor
6	BJB (battery junction box)
7	EJB (engine junction box)
8	TSS (tandem solenoid starter) Motor
9	Terrain Response® Switchpack
10	IC (instrument cluster)
11	KVM (keyless vehicle module)
12	Connection to Other Control Modules on MS (medium speed) CAN Body Bus
13	Connection to Other Control Modules on HS (high speed) CAN Powertrain Bus
14	GWM (gateway module)
15	TCS (transmission control switch)
16	CJB (central junction box)
17	Stop/Start Switch
18	ECM (engine control module)
19	AAT (ambient air temperature) Sensor
20	Accelerator Pedal
21	Brake Pedal Switch